UG" END SEMESTER CLASS TEST – DECEMBER 2023
(REGULAR-ODD SEMESTER)
1ST "YEAR (1ST SEMESTER) - MECHANICAL ENGINEERING
ENGINEERING MECHANICS
MENUGES01
For ECE & EEN students

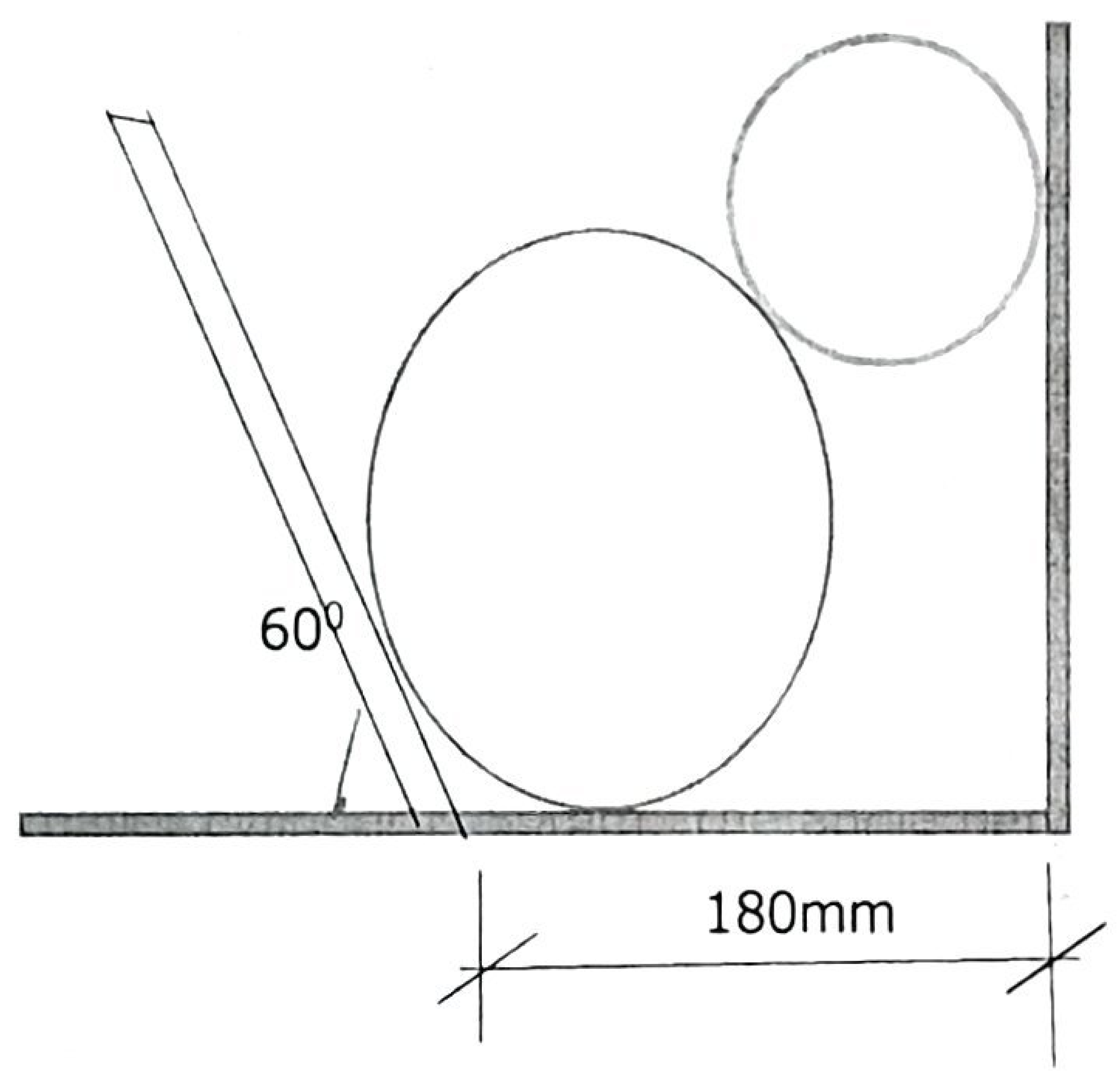
(Full Marks – 50, Time – 1&1/2 Hours)

(Assume missing data if any)

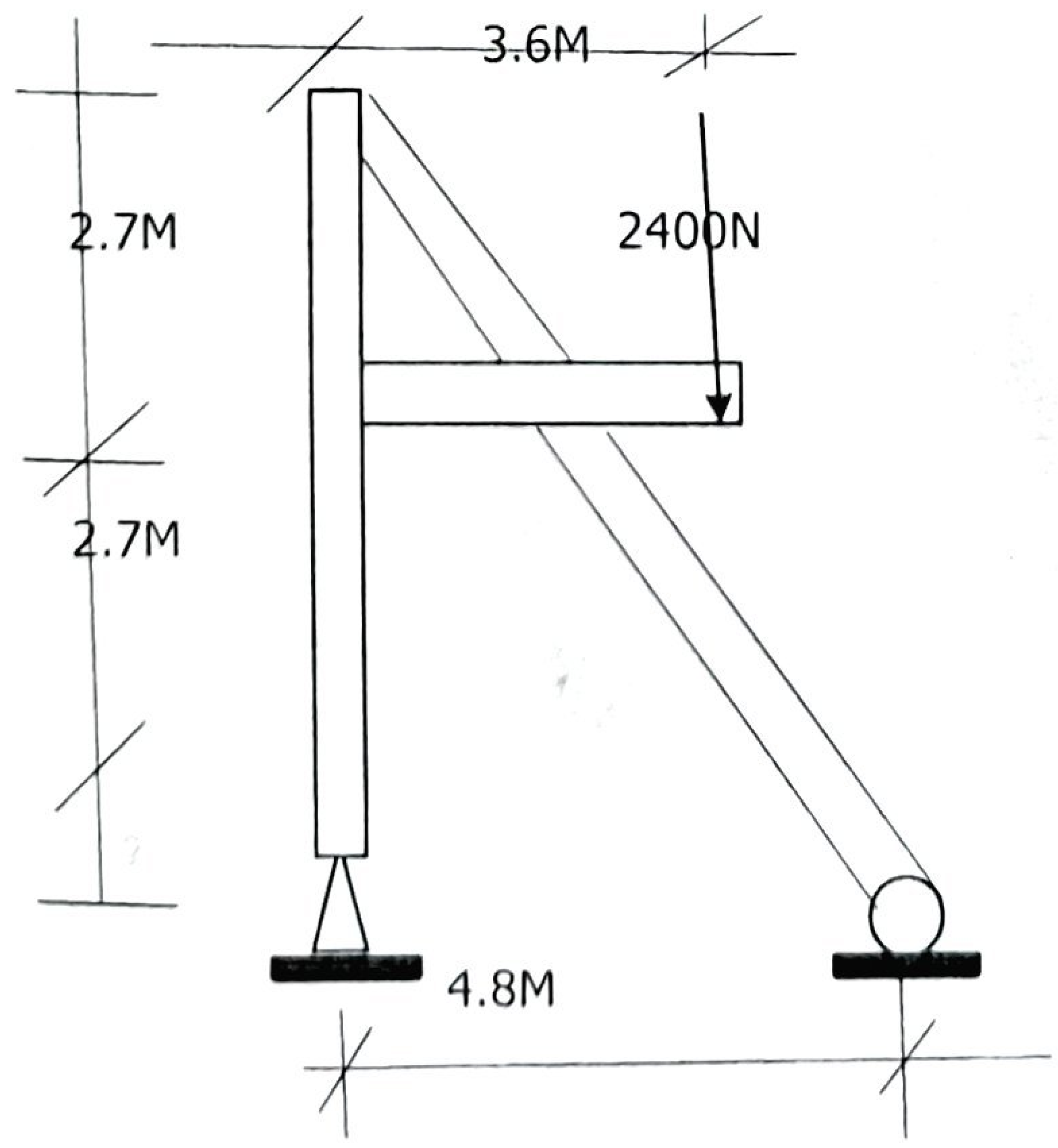
Group- A.

(Answer any three questions) [3x (10) =30]

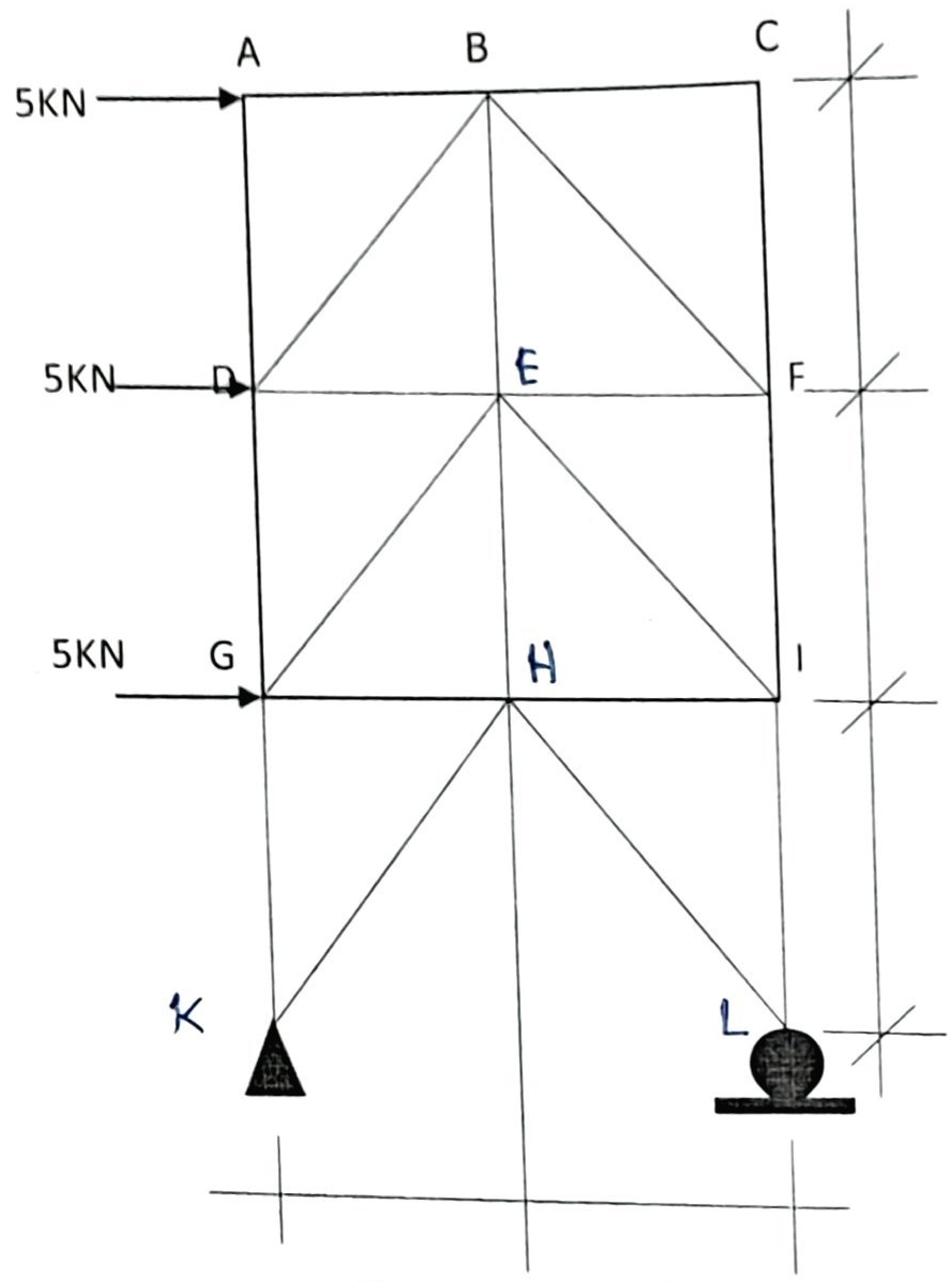
1. Two smooth cylinders each of weight 2000N and 800N respectively rest on a horizontal channel having one inclined angle and one vertical wall as shown in the figure. The radius of bigger cylinder is 100mm while that of smaller cylinder is 50mm. The angle between inclined wall with horizontal is 60^0. Find the pressure exerted on the walls and floors at point of contact A, B, C & D. [10]



2. (a) Determine the component of the forces acting on each of the frame shown in figure.

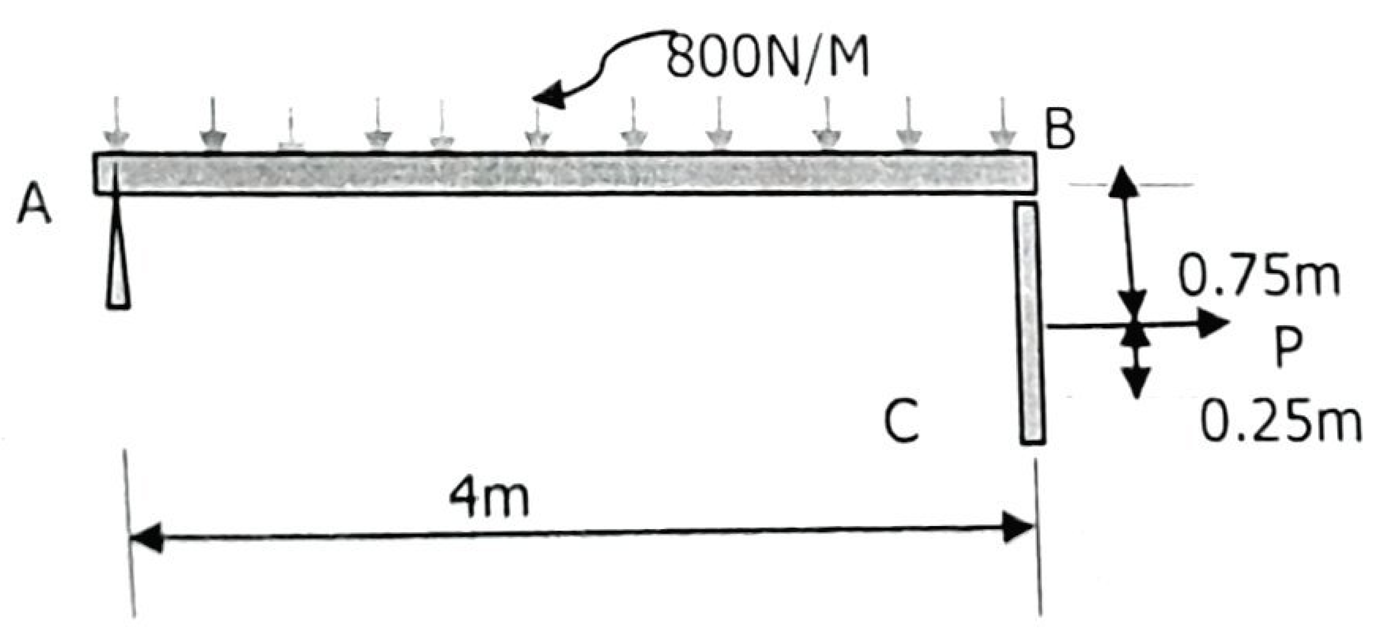


(b) Find the axial force in member DG of truss shown below. [5+5}



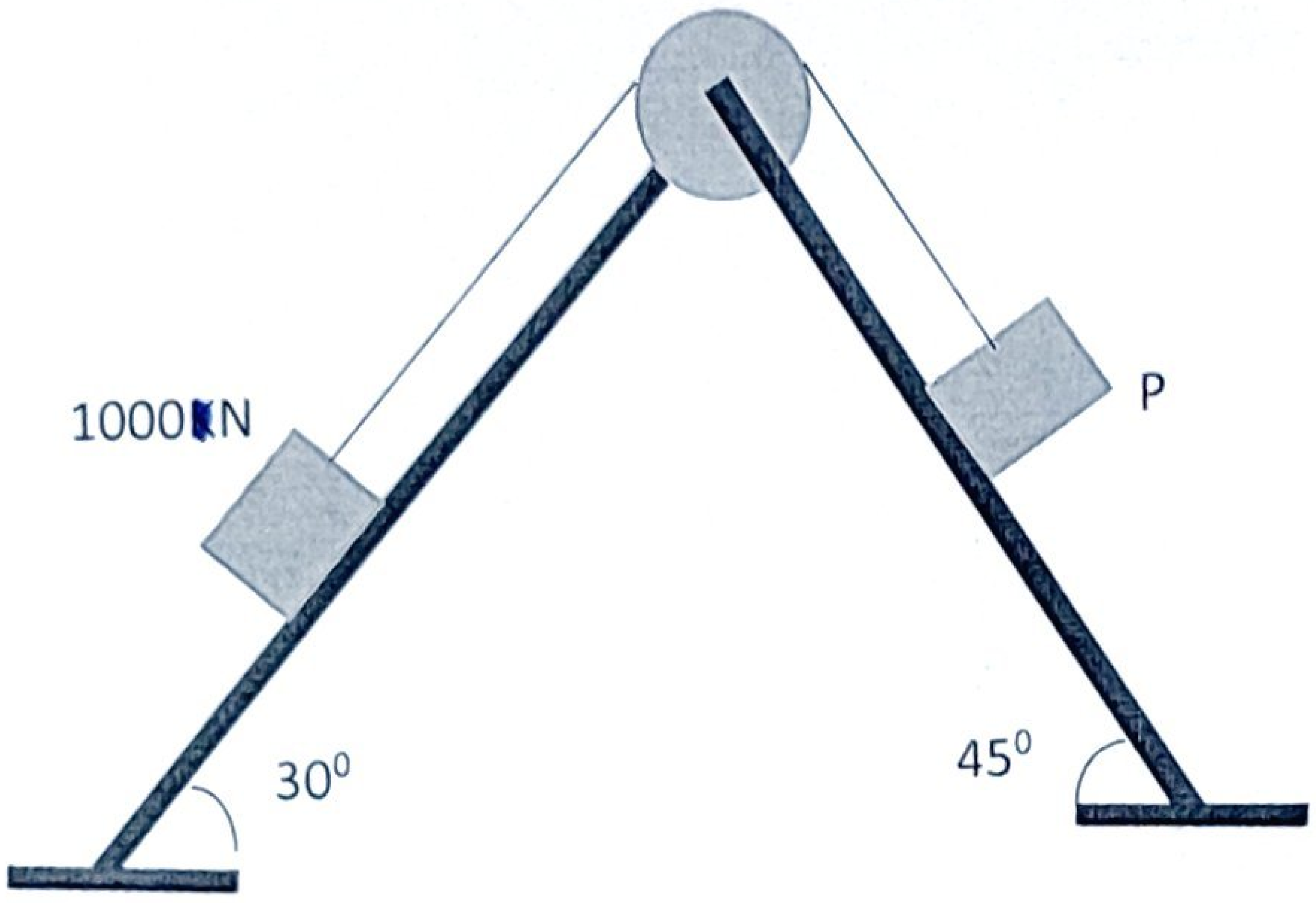
3. (a) Define friction, angle of repose and angle of friction. Establish relation between angle of repose and angle of friction. State principle of equilibrium.

(b) Beam AB is subjected to a uniform load 200N/m and is supported by at B by post BC as shown in Figure. If the co-efficient of static friction at B and C are μ_B 0.2 and μ_C 0.5, determine the force P needed to pull the post out from the under the beam. Neglect the weight of members and thickness of the beam. [5+5]



4. (a) Define centroid, center of mass and center of gravity. In a composite body will the centre of mass and centroid in same position?

(b) A weight of 1000N resting over a smooth surface inclined at 30^0 with horizontal, is supported by an effort (P) resting on a smooth surface inclined 45^0 with horizontal. By using principle of virtual work, calculate the value of effort 'P'. [5+5]



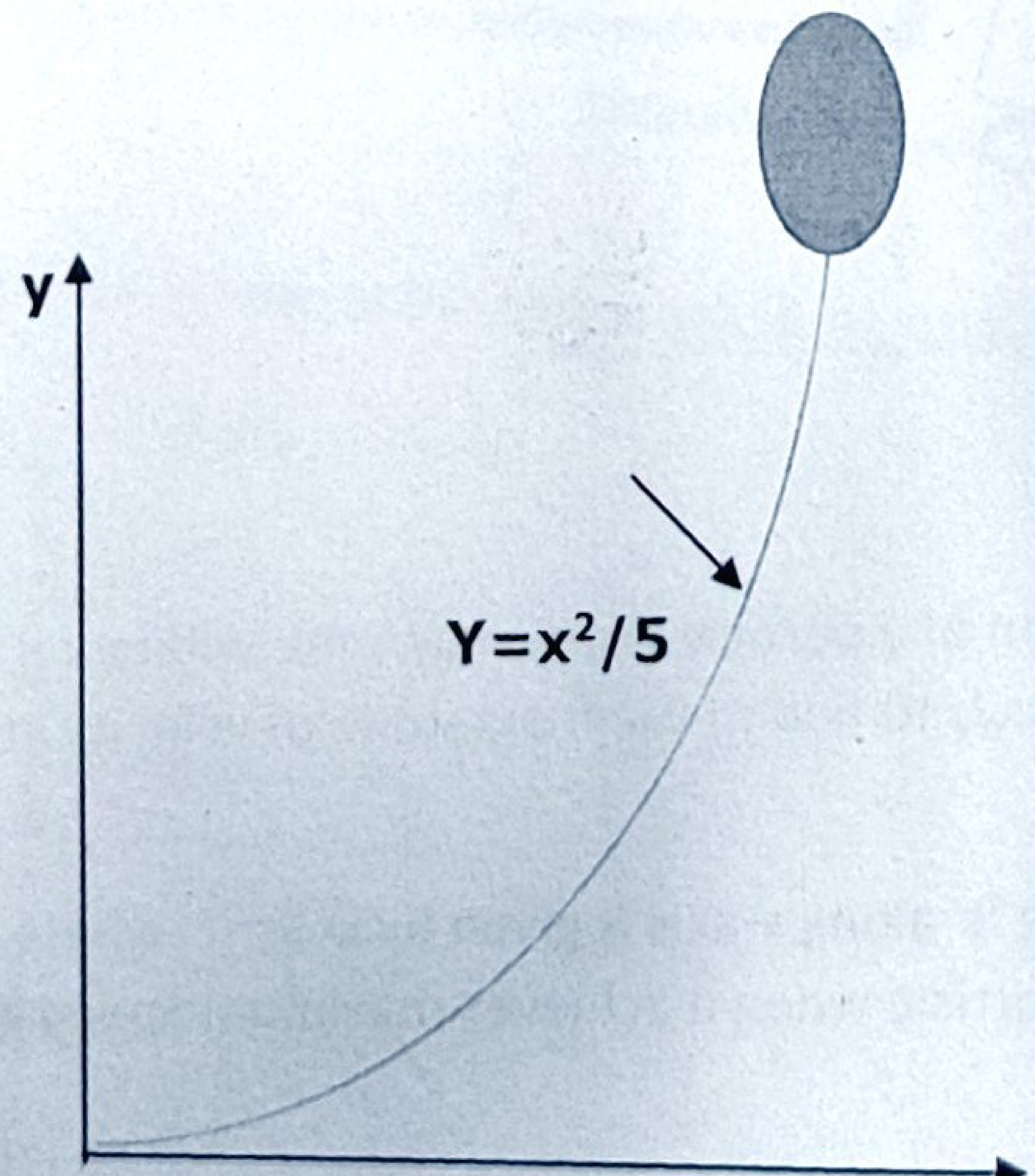
Group –B

Answer any two questions. [2x10=20]

1 (a). At any instant the horizontal position of weather balloon in figure is defined by $x = (2t)$ m, where t is in second. If the equation of the path is $y = x^2/5$, determine the magnitude and direction of

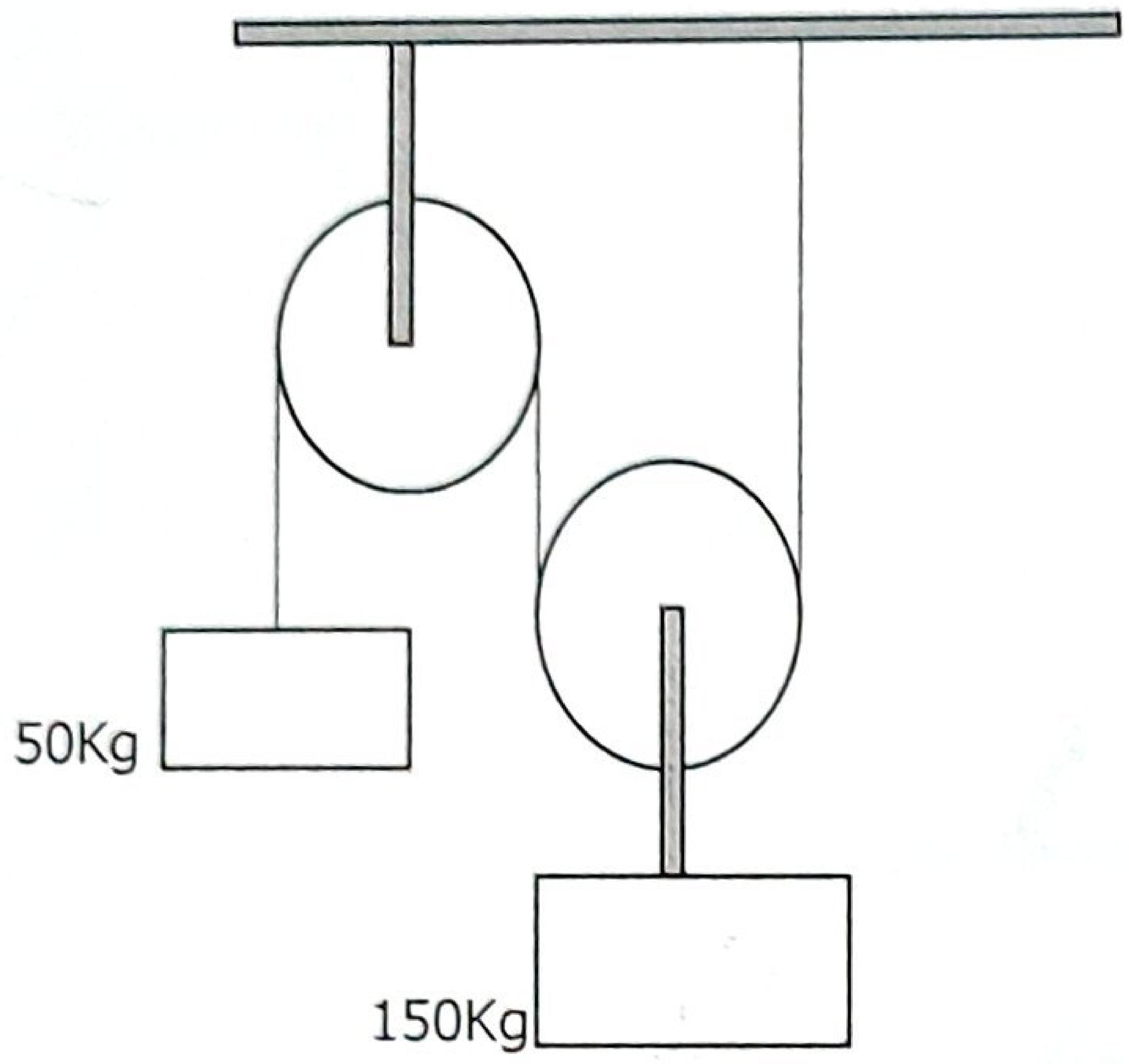
(i) The velocity and (ii) acceleration when t= 2 sec.

(ii) Acceleration when t= 2 sec.

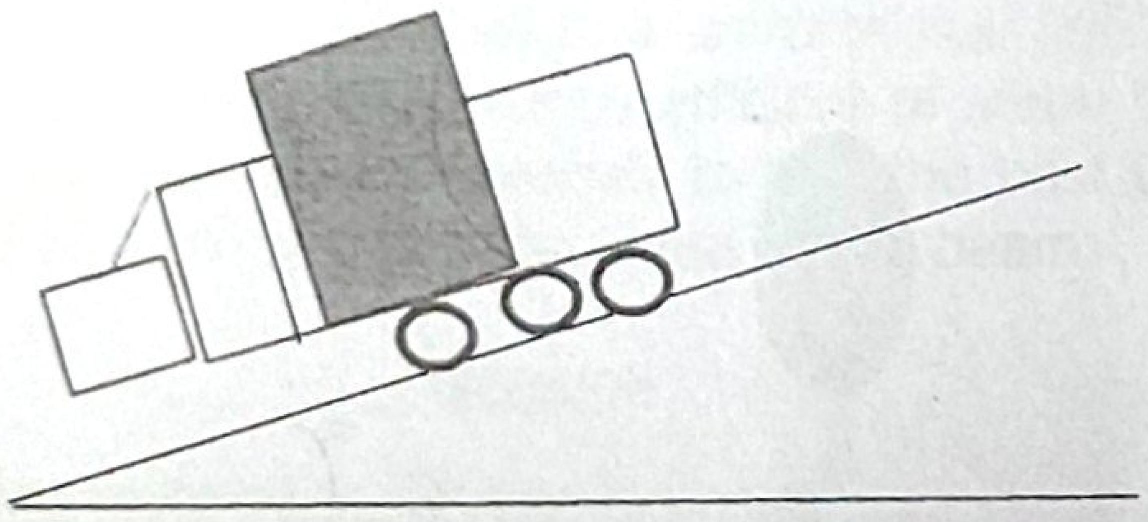


1 (b) An elevator of mass 3000kg is moving downward vertically with a constant acceleration starting from the rest. It travels a distance of 40m during travel in interval of 10sec. Find the cable tension in Newton during this time. Assume, $g = 10 m/s^2$. [5+5]

2. (a) Determine the tension in the strings and acceleration of two blocks of mass 50Kg and 150Kg connected by a string and a frictionless & weightless pulley as shown in figure. Assume $g=10m/s^2$



(b) A trailer is travelling down with 2^0 inclination with horizon (shown in figure) at a speed of 25m/s, when the brakes are applied. Knowing the static and kinetic coefficient of frictions between the loads and flatbed trailer shown are 0.4 and 0.35 respectively. Determine the shortest time in which the trailer brought to a stop if the load is not to shift. [5+5]



3. (a) At an instant, the horizontal position of a particle is given by x=8t, where x is in metre & t is in sec. If the equation of path is given by y=x2/10 find (i) the magnitude of velocity in m/s and (ii) acceleration in m/s^2 at t=2sec.

(b) The position of a particle w.r.t time 't' along x-axis is given by $x=9t^2-t^3$ where x is in metre and t in sec. What will be the position of the particle when it achieves maximum Speed along the positive direction of x axis? [5+5]

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